

Area (Part I)

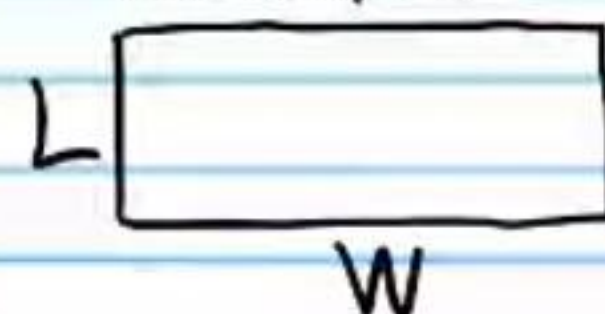
Name
Date
Course

Area of a Rectangle

Example:

$$A = LW$$

L = length & W = width

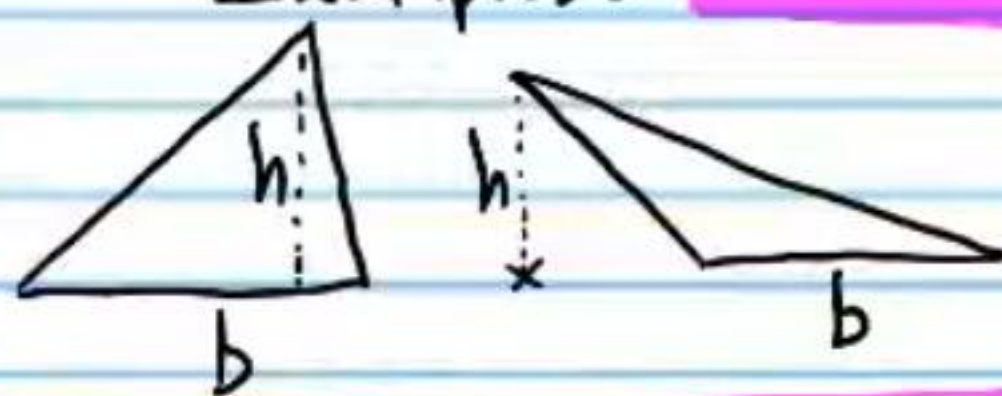


Area of a Triangle

Examples:

$$A = \frac{1}{2}bh$$

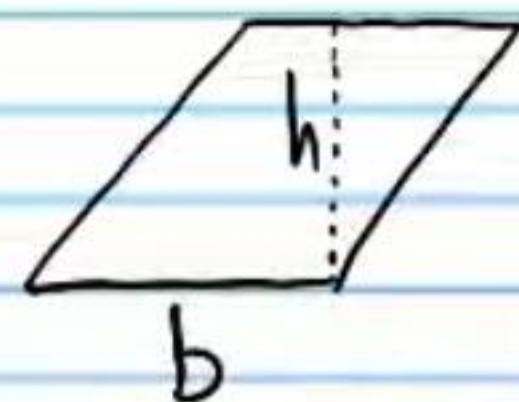
b = base & h = height



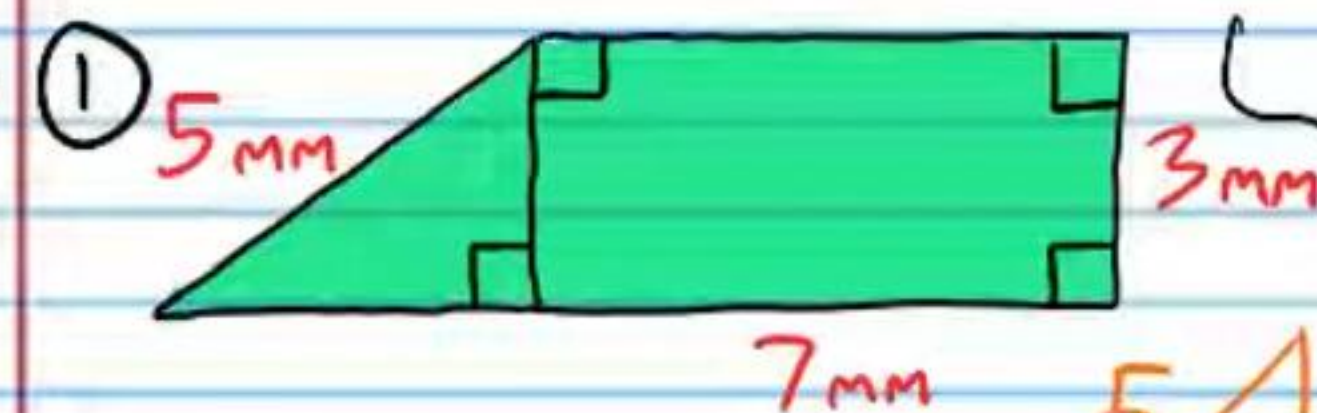
Area of a Parallelogram

$$A = bh$$

b = base & h = height



Find the area of each shaded region below.

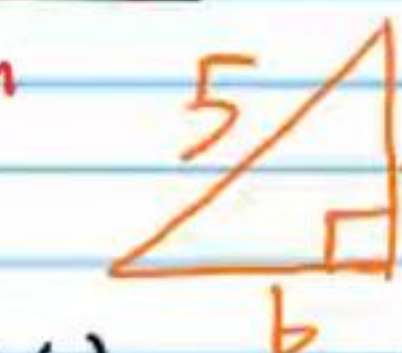


$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{Tri}}$$

$$A_{\text{Tot}} = 7(3) + \frac{1}{2}(4)(3)$$

$$A_{\text{Tot}} = 21 + 6$$

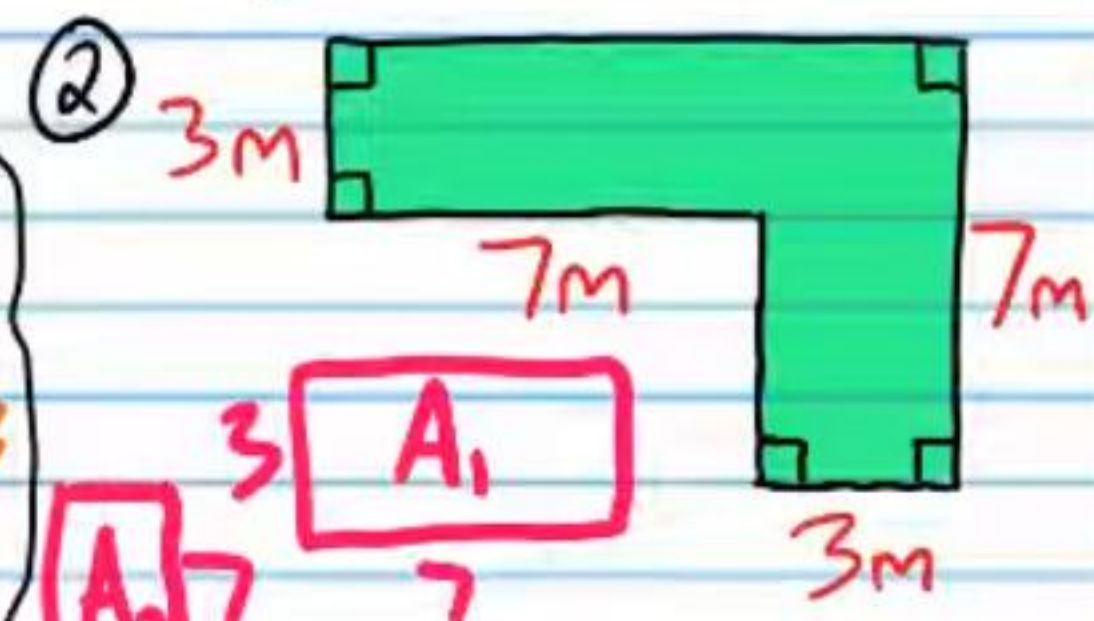
$$A_{\text{Tot}} = 27 \text{ mm}^2$$



$$a^2 + b^2 = c^2$$

$$3^2 + b^2 = 5^2$$

$$b = 4$$

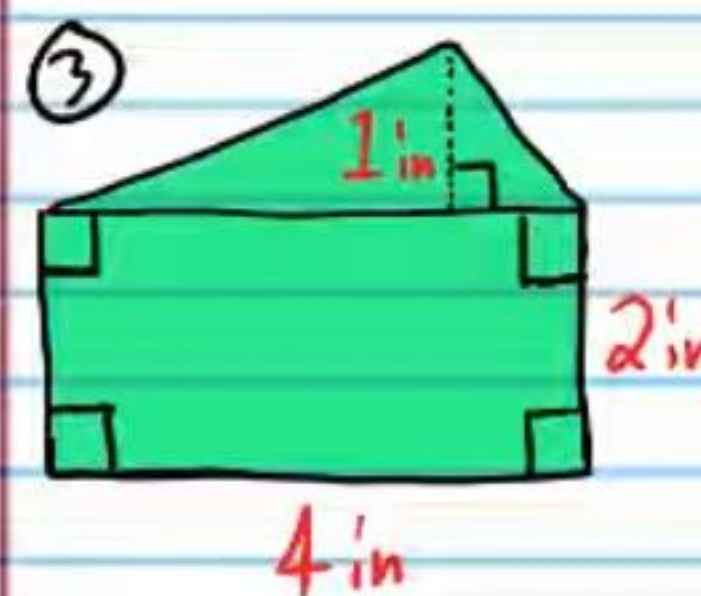


$$A_{\text{Tot}} = A_{\text{Rect1}} + A_{\text{Rect2}}$$

$$A_{\text{Tot}} = 7(3) + 7(3)$$

$$A_{\text{Tot}} = 21 + 21$$

$$A_{\text{Tot}} = 42 \text{ m}^2$$

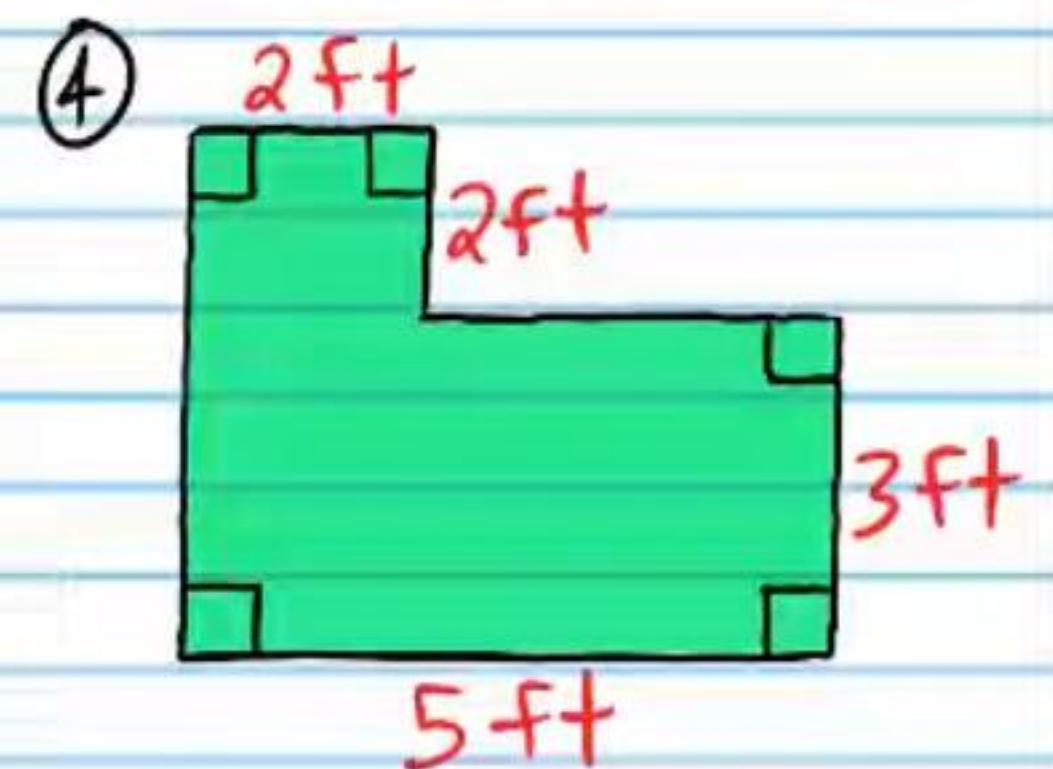


$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{Tri}}$$

$$A_{\text{Tot}} = 4(2) + \frac{1}{2}(4)(1)$$

$$A_{\text{Tot}} = 8 + 2$$

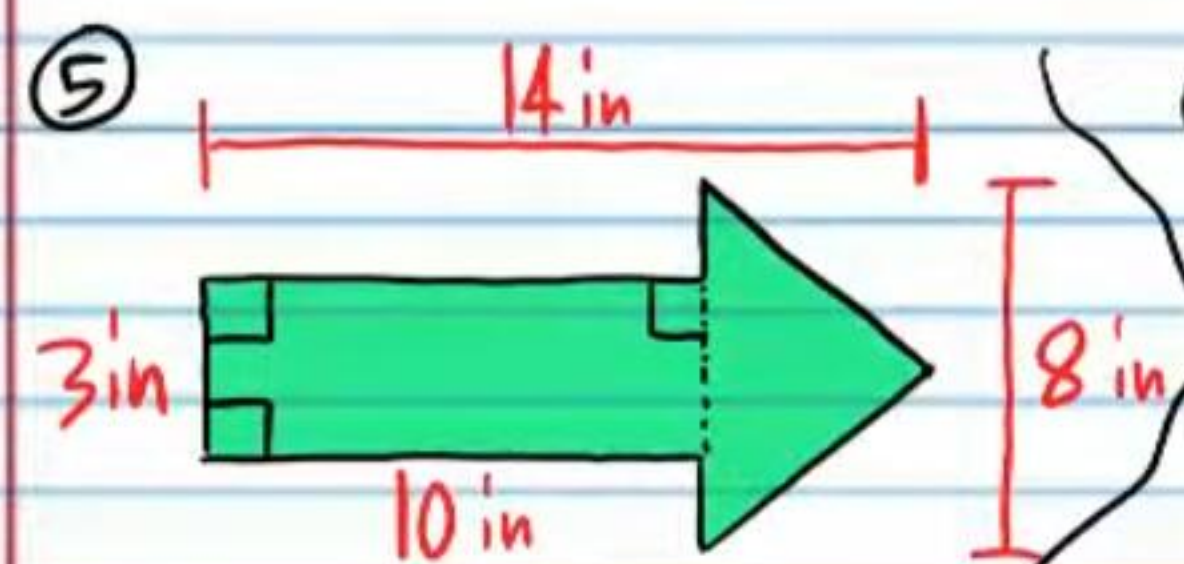
$$A_{\text{Tot}} = 10 \text{ in}^2$$



$$A_{\text{Tot}} = 5(3) + 2(2)$$

$$A_{\text{Tot}} = 15 + 4$$

$$A_{\text{Tot}} = 19 \text{ ft}^2$$



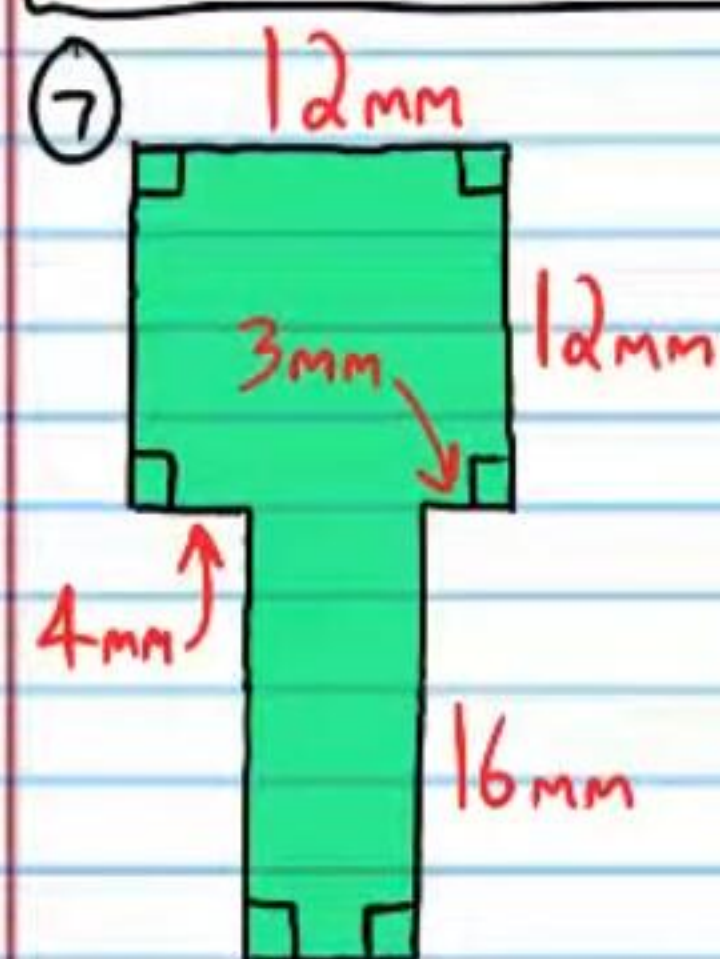
$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{Tri}}$$

$$A_{\text{Tot}} = 3(10) + \frac{1}{2}(8)(14-10)$$

$$A_{\text{Tot}} = 30 + \frac{1}{2}(8)(4)$$

$$A_{\text{Tot}} = 30 + 16$$

$$A_{\text{Tot}} = 46 \text{ in}^2$$

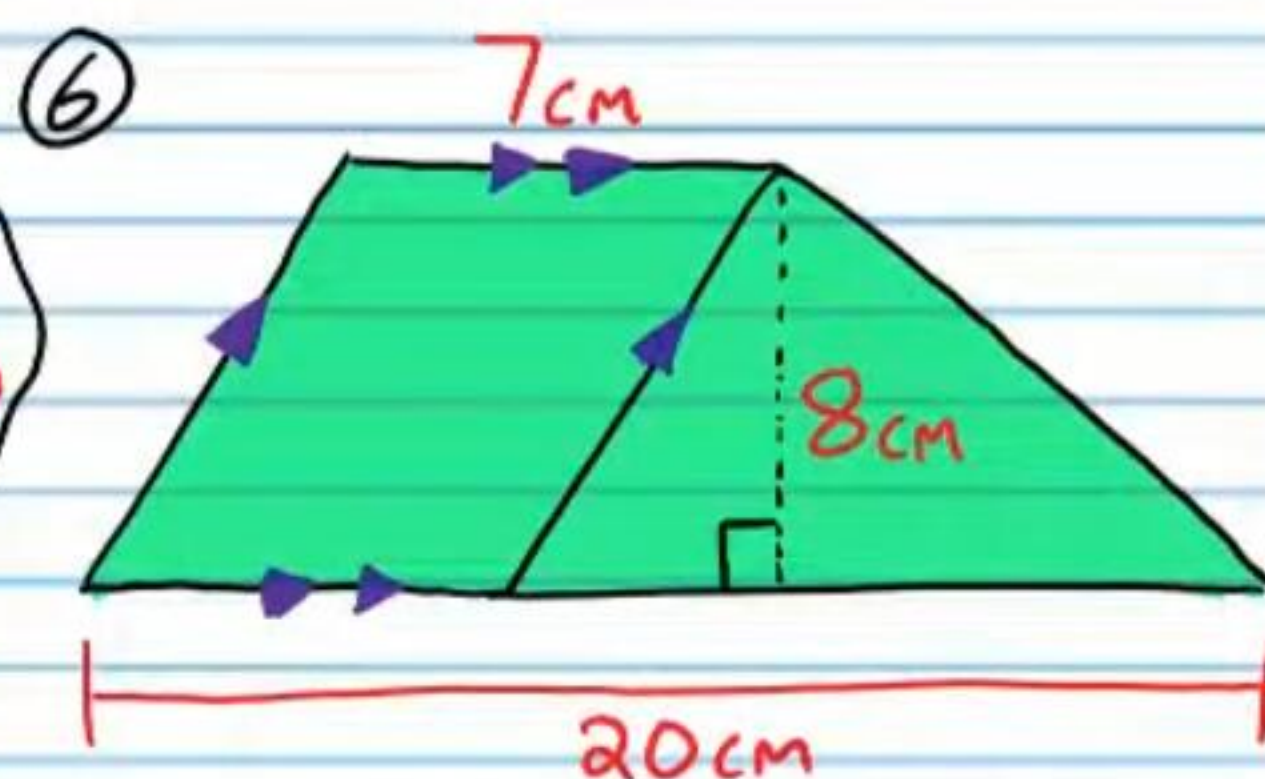


$$A_{\text{Tot}} = A_{\text{Rect1}} + A_{\text{Rect2}}$$

$$A_{\text{Tot}} = 12(12) + 16(12-4-3)$$

$$A_{\text{Tot}} = 144 + 16(5)$$

$$A_{\text{Tot}} = 224 \text{ mm}^2$$

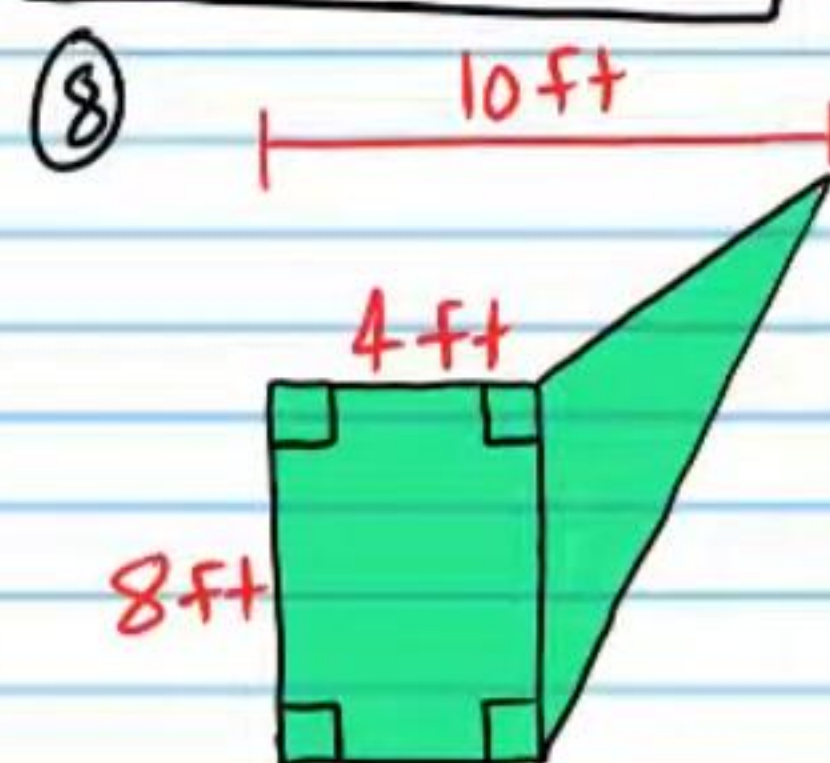


$$A_{\text{Tot}} = A_{\text{Para}} + A_{\text{Tri}}$$

$$A_{\text{Tot}} = 7(8) + \frac{1}{2}(20-7)(8)$$

$$A_{\text{Tot}} = 56 + \frac{1}{2}(13)(8)$$

$$A_{\text{Tot}} = 108 \text{ cm}^2$$

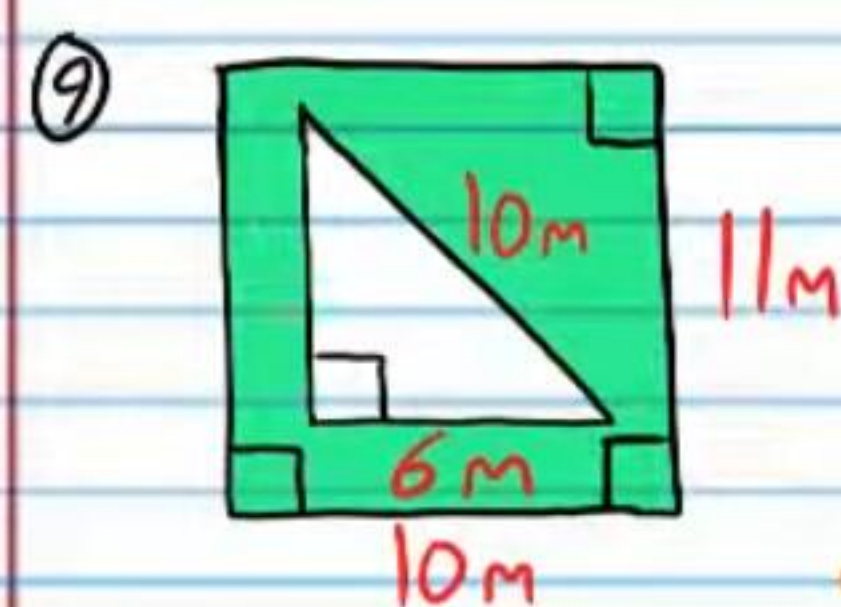


$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{Tri}}$$

$$A_{\text{Tot}} = 4(8) + \frac{1}{2}(8)(10-4)$$

$$A_{\text{Tot}} = 32 + \frac{1}{2}(8)(6)$$

$$A_{\text{Tot}} = 56 \text{ ft}^2$$

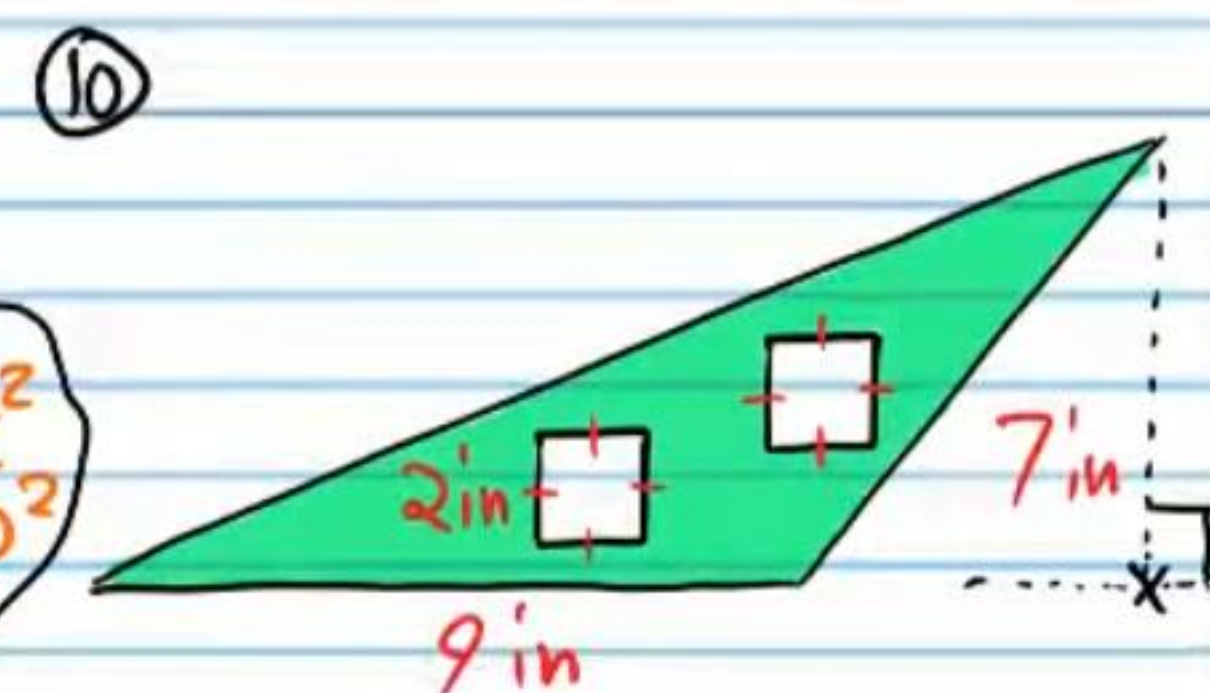


$$A_{\text{Tot}} = A_{\text{Rect}} - A_{\text{Tri}}$$

$$A_{\text{Tot}} = 10(11) - \frac{1}{2}(6)(8)$$

$$A_{\text{Tot}} = 110 - 24$$

$$A_{\text{Tot}} = 86 \text{ m}^2$$



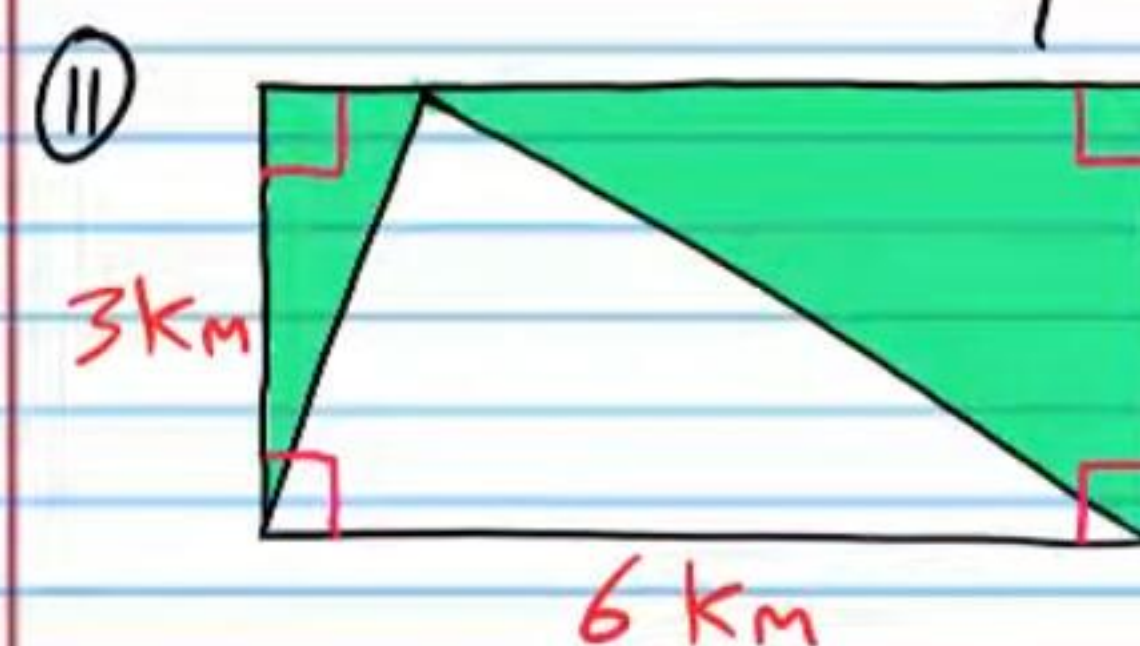
$$A_{\text{Tot}} = A_{\text{Tri}} - A_{\text{Squ}}$$

$$A_{\text{Tot}} = \frac{1}{2}(9)(7) - (2(2) + 2(2))$$

$$A_{\text{Tot}} = 31.5 - (4 + 4)$$

$$A_{\text{Tot}} = 31.5 - 8$$

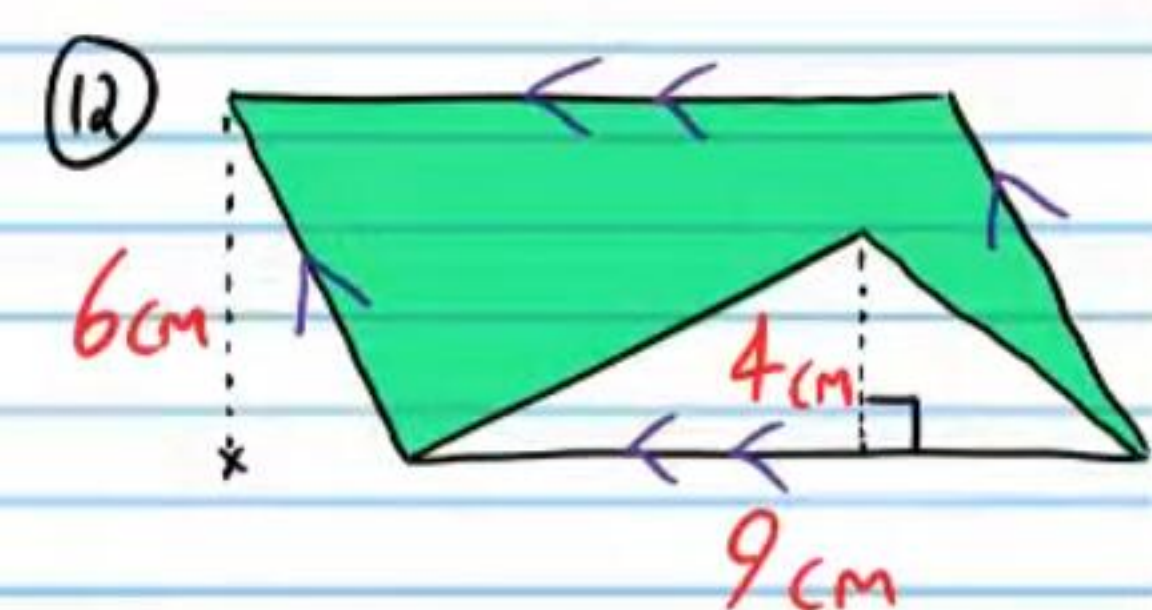
$$A_{\text{Tot}} = 23.5 \text{ in}^2$$



$$A_{\text{Tot}} = A_{\text{Rect}} - A_{\text{Tri}}$$

$$A_{\text{Tot}} = 3(6) - \frac{1}{2}(6)(3)$$

$$A_{\text{Tot}} = 9 \text{ km}^2$$



$$A_{\text{Tot}} = A_{\text{Para}} - A_{\text{Tri}}$$

$$A_{\text{Tot}} = 9(6) - \frac{1}{2}(9)(4)$$

$$A_{\text{Tot}} = 54 - 18$$

$$A_{\text{Tot}} = 36 \text{ cm}^2$$

Area of a Circle

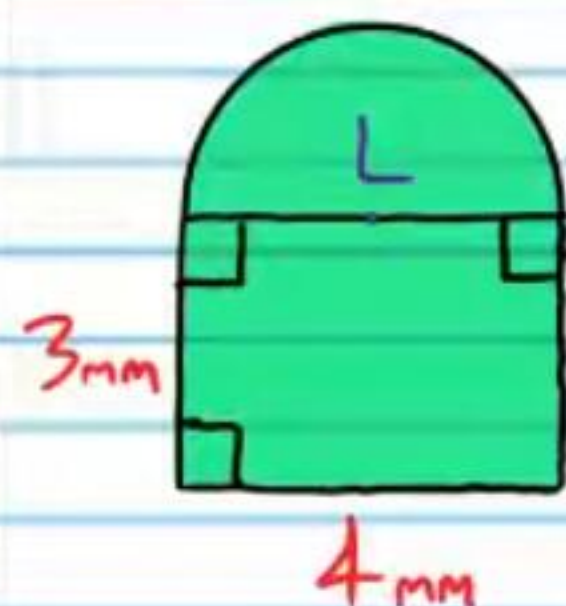
$$A = \pi r^2$$

$r = \text{radius}$



Find the area of each shaded region below.

(13) L is the center of the semicircle.



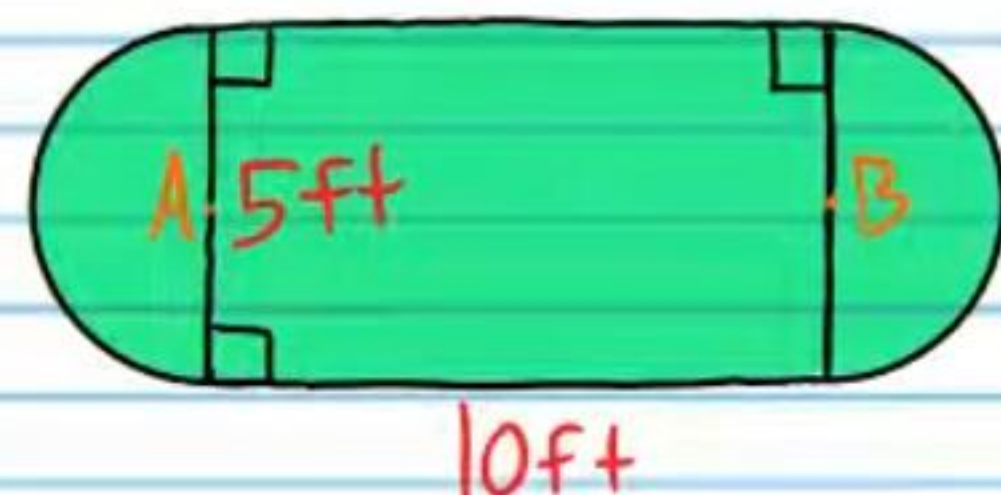
$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{semicircle}}$$

$$A_{\text{Tot}} = 3(4) + \frac{\pi(2)^2}{2}$$

$$A_{\text{Tot}} = 12 + \frac{\pi(4)}{2}$$

$$A_{\text{Tot}} = 12 + 2\pi \text{ mm}^2$$

(14) A & B are centers of the semicircles.



$$A_{\text{Tot}} = A_{\text{Rect}} + A_{\text{semicircles}}$$

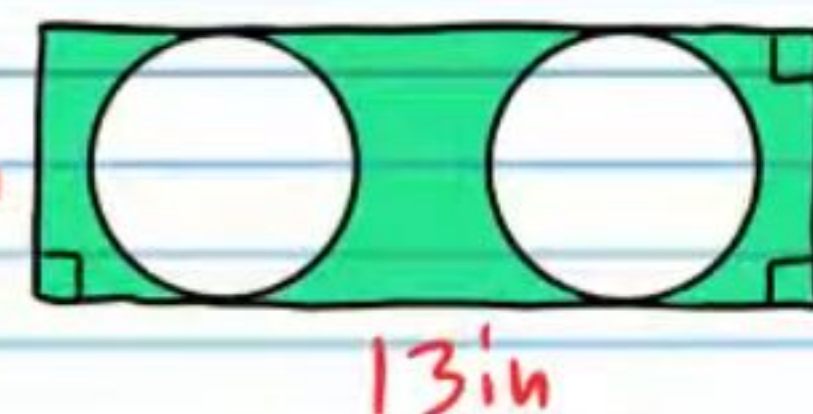
$$A_{\text{Tot}} = 5(10) + \frac{\pi(\frac{5}{2})^2}{2} + \frac{\pi(\frac{5}{2})^2}{2}$$

$$A_{\text{Tot}} = 50 + \frac{\pi(6.25)}{2} + \frac{\pi(6.25)}{2}$$

$$A_{\text{Tot}} = 50 + 6.25\pi$$

$$A_{\text{Tot}} \approx 69.63 \text{ ft}^2$$

(15)



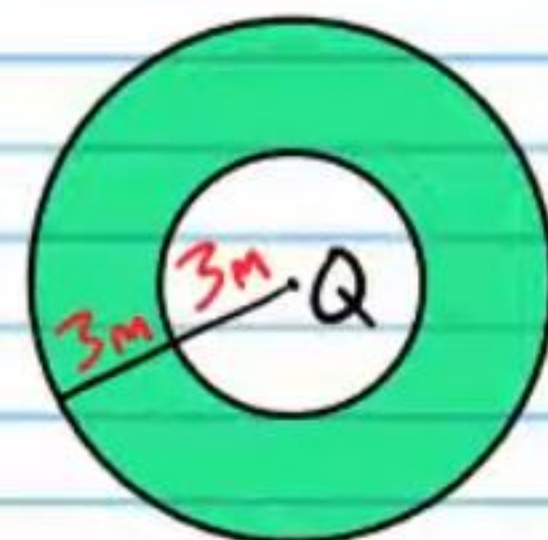
$$A_{\text{Tot}} = A_{\text{Rect}} - A_{\text{circles}}$$

$$A_{\text{Tot}} = 3(13) - \left[\pi\left(\frac{3}{2}\right)^2 + \pi\left(\frac{3}{2}\right)^2 \right]$$

$$A_{\text{Tot}} = 39 - [4.5\pi]$$

$$A_{\text{Tot}} \approx 24.86 \text{ in}^2$$

(16) Q is the center.



$$A_{\text{Tot}} = A_{\text{circle1}} - A_{\text{circle2}}$$

$$A_{\text{Tot}} = \pi(3+3)^2 - \pi(3)^2$$

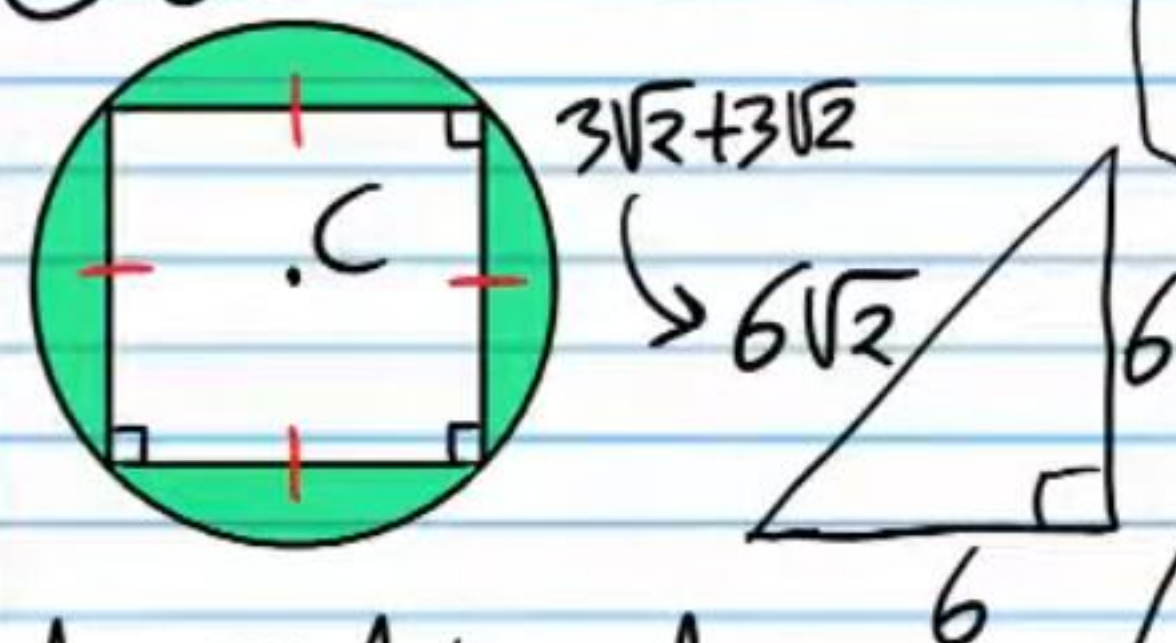
$$A_{\text{Tot}} = \pi(6)^2 - 9\pi$$

$$A_{\text{Tot}} = 36\pi - 9\pi$$

$$A_{\text{Tot}} = 27\pi \text{ m}^2$$

$$A_{\text{Tot}} \approx 84.82 \text{ m}^2$$

(17) $\odot C$ has radius $3\sqrt{2}$ in. (18) M is the center.

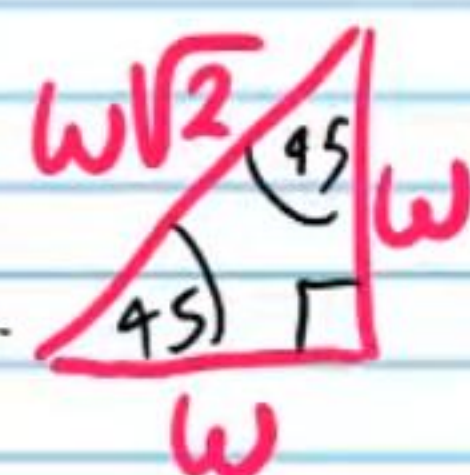
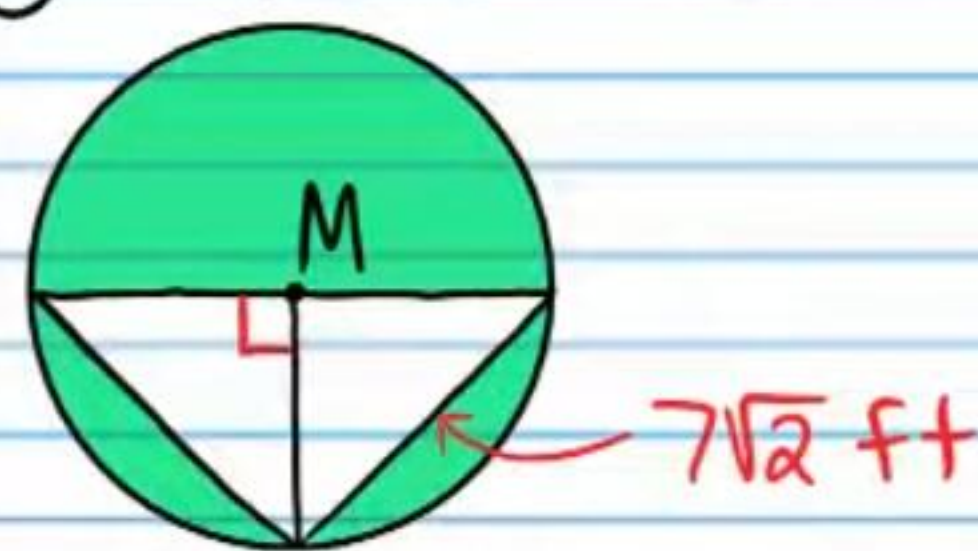


$$A_{\text{Tot}} = A_{\text{circle}} - A_{\text{sqn}}$$

$$A_{\text{Tot}} = \pi(3\sqrt{2})^2 - 6(6)$$

$$A_{\text{Tot}} = 18\pi - 36 \text{ in}^2$$

$$A_{\text{Tot}} \approx 20.548 \text{ in}^2$$



$$A_{\text{Tot}} = A_{\text{circle}} - A_{\text{tri}}$$

$$A_{\text{Tot}} = \pi(7)^2 - \frac{1}{2}(7+7)(7)$$

$$A_{\text{Tot}} = 49\pi - \frac{1}{2}(14)(7)$$

$$A_{\text{Tot}} = 49\pi - 7(7)$$

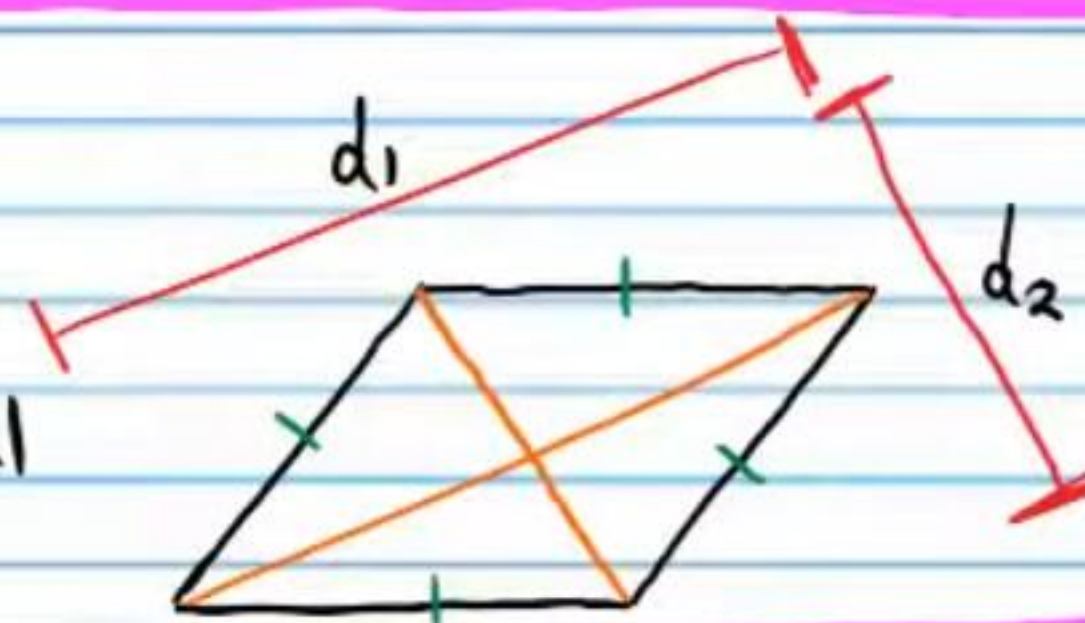
$$A_{\text{Tot}} = 49\pi - 49 \text{ ft}^2$$

$$A_{\text{Tot}} \approx 104.9 \text{ ft}^2$$

Area of a Rhombus

$$A = \frac{1}{2} d_1 d_2$$

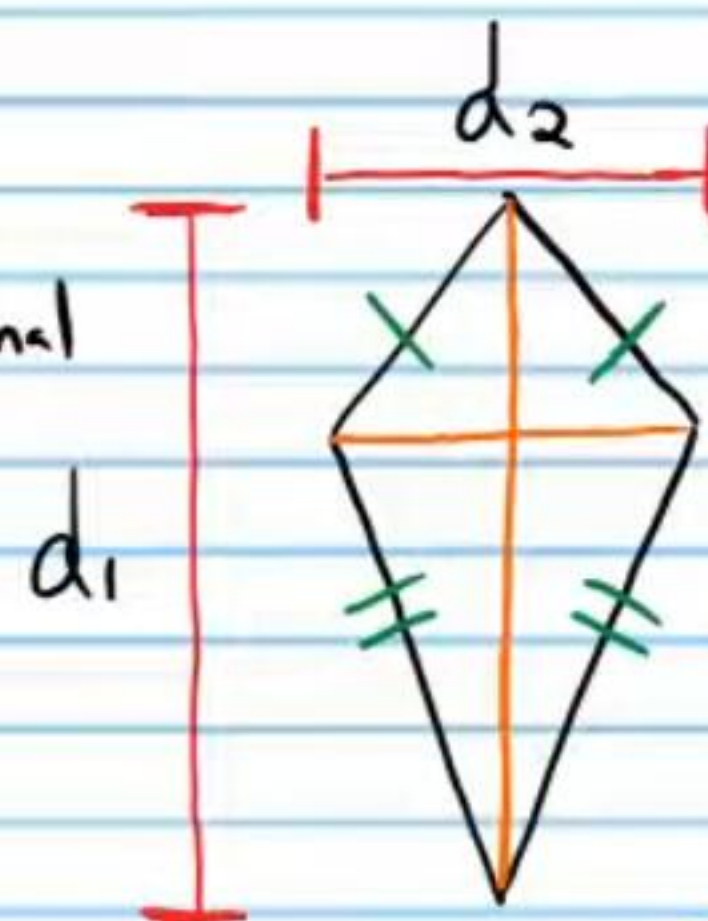
$d_1 = \text{diagonal} \ \& \ d_2 = \text{diagonal}$



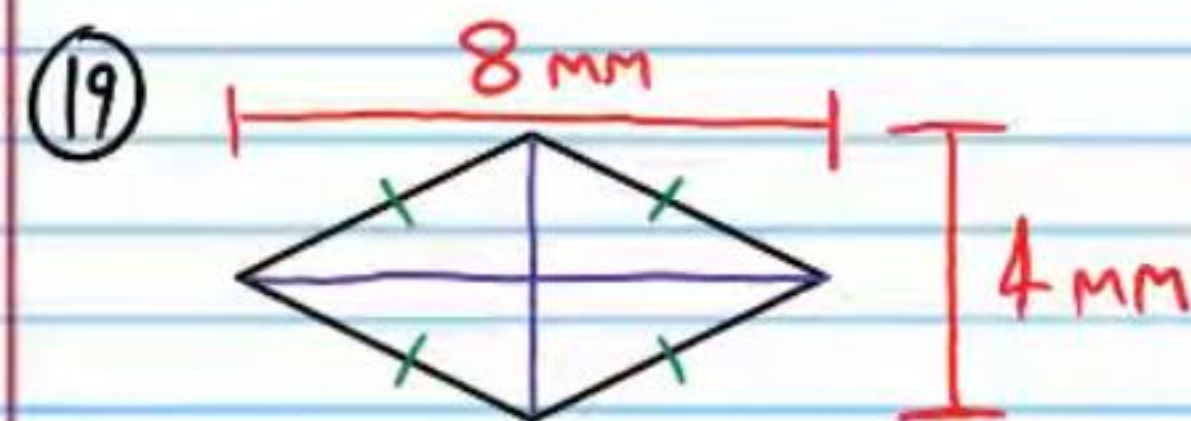
Area of a kite

$$A = \frac{1}{2} d_1 d_2$$

$d_1 = \text{diagonal} \ \& \ d_2 = \text{diagonal}$



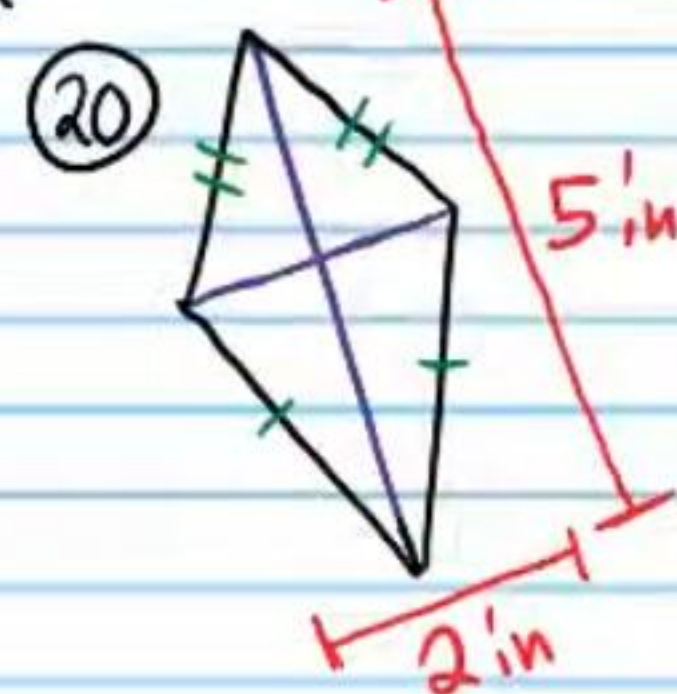
Find the area of each quadrilateral below.



$$A = \frac{1}{2}(8)(4)$$

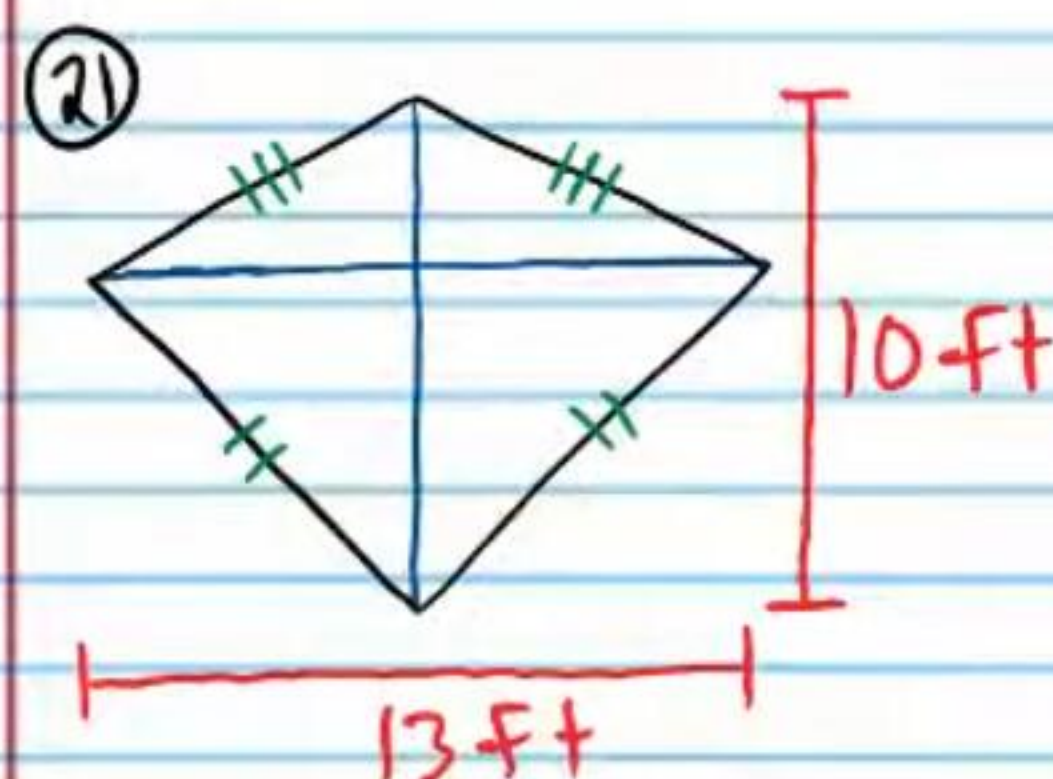
$$A = \frac{1}{2}(32)$$

$$A = 16 \text{ mm}^2$$



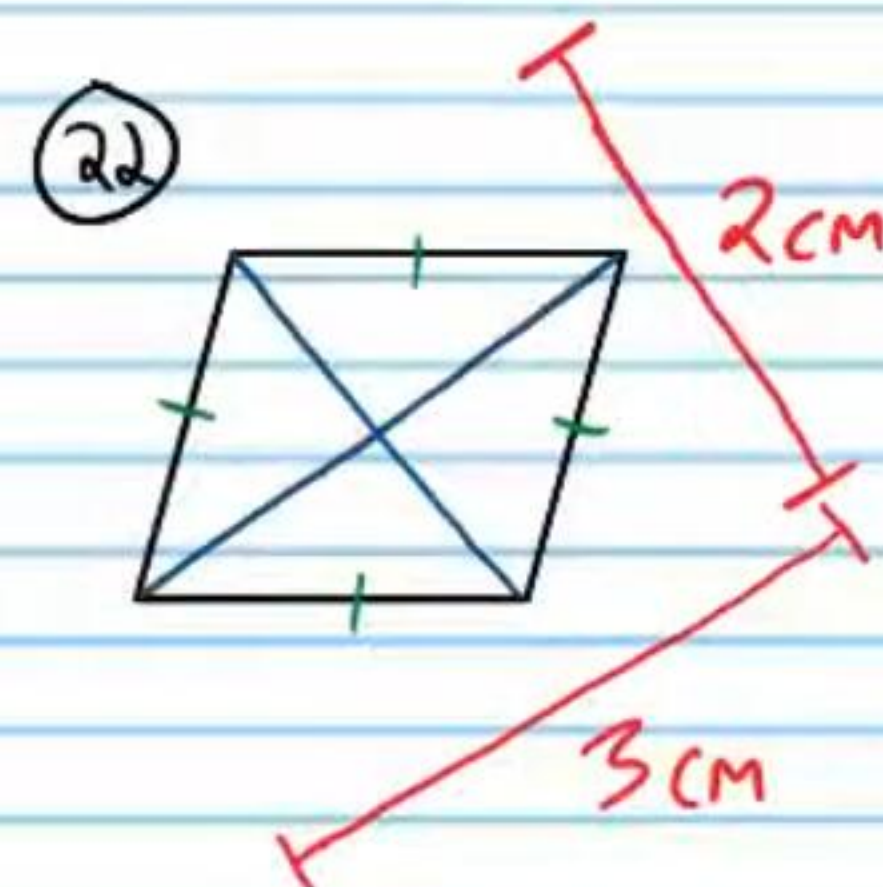
$$A = \frac{1}{2}(5)(2)$$

$$A = 5 \text{ in}^2$$



$$A = \frac{1}{2}(13)(10)$$

$$A = 65 \text{ ft}^2$$

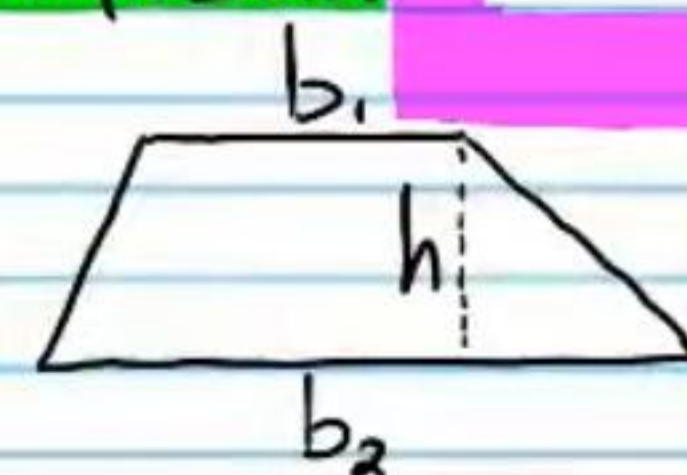


$$A = \frac{1}{2}(3)(2)$$

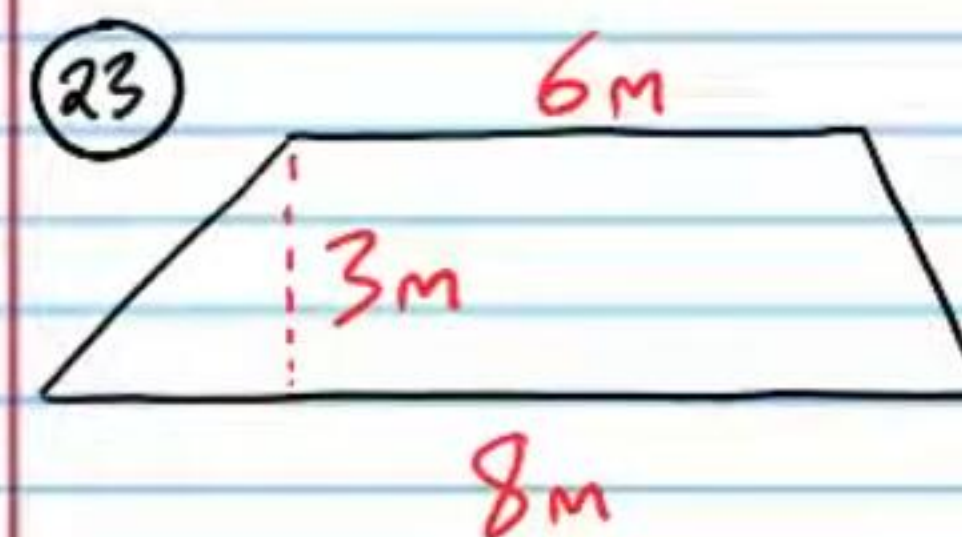
$$A = 3 \text{ cm}^2$$

Area of a Trapezoid

$$A = \frac{h(b_1 + b_2)}{2}$$



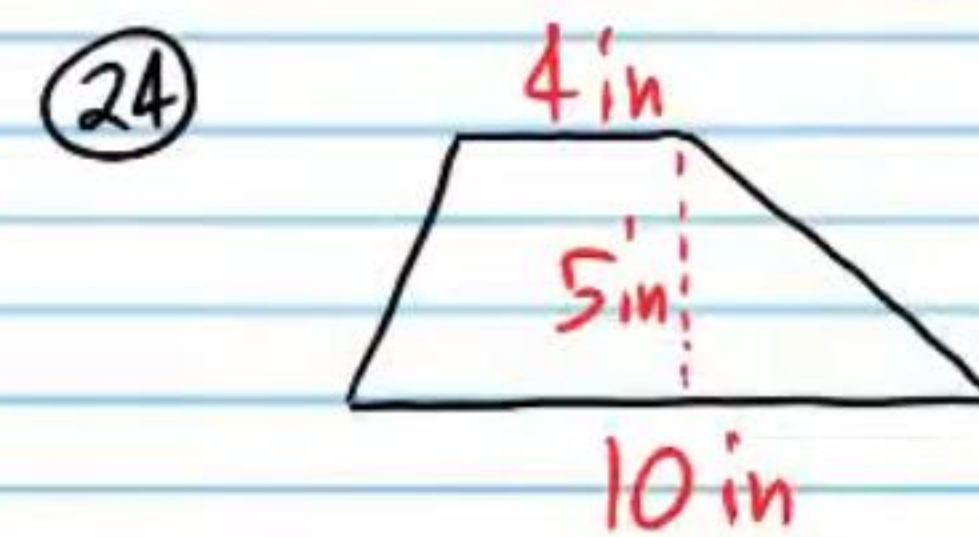
Find the area of each trapezoid below.



$$A = \frac{3(6+8)}{2}$$

$$A = \frac{3(14)}{2}$$

$$A = 21 \text{ m}^2$$



$$A = \frac{5(10+4)}{2}$$

$$A = \frac{5(14)}{2}$$

$$A = 35 \text{ in}^2$$

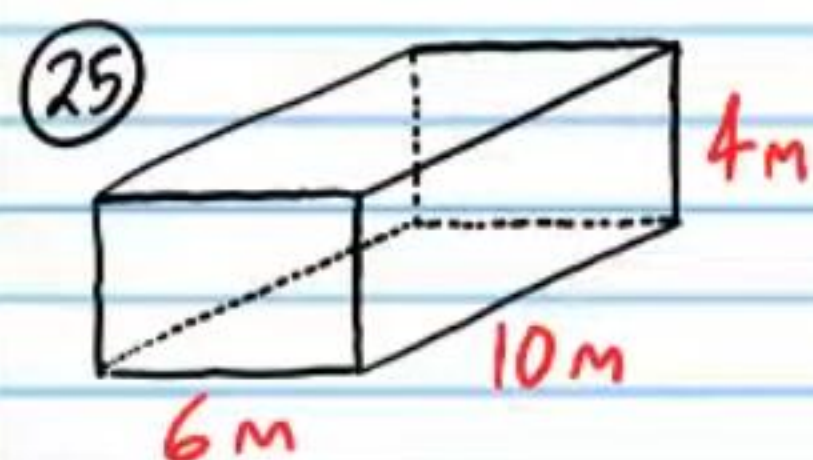
Surface Area

Prism: A three dimensional shape with two congruent and parallel sides called bases.

Rectangular Prism: A three dimensional shape with six rectangular sides.



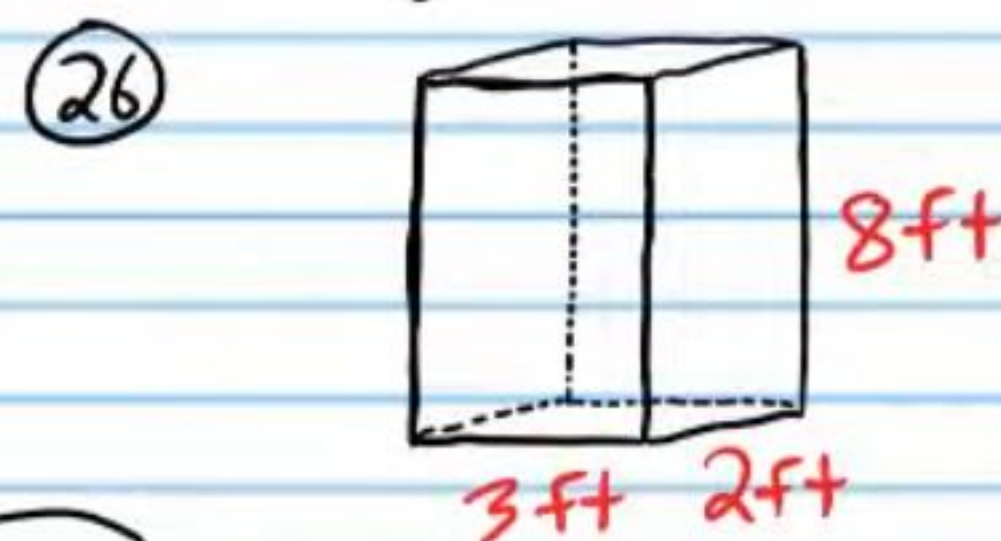
Find the surface area of each rectangular prism below.



$$A = 10(4)(2) + 6(4)(2) + 6(10)(2)$$

$$A = 80 + 48 + 120$$

$$A = 248 \text{ m}^2$$

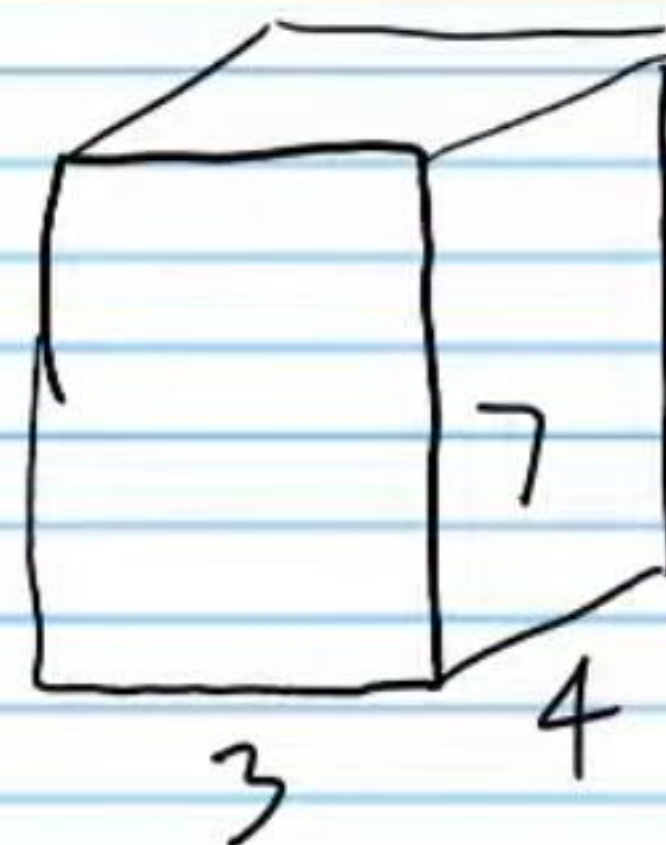


$$A = 2(8)(2) + 3(8)(2) + 3(2)(2)$$

$$A = 32 + 48 + 12$$

$$A = 92 \text{ ft}^2$$

(27) If a refrigerator is in the shape of a rectangular prism with dimensions $3 \text{ ft} \times 4 \text{ ft} \times 7 \text{ ft}$, find its surface area.

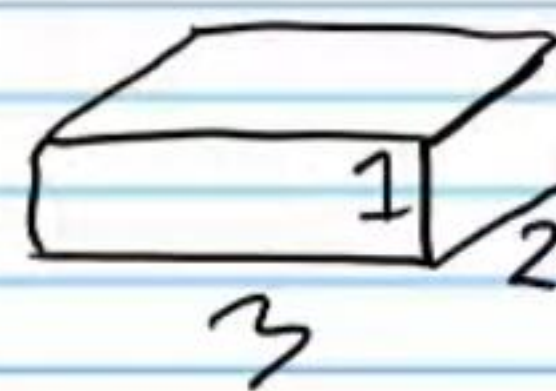


$$A = 3(7)(2) + 7(4)(2) + 3(4)(2)$$

$$A = 42 + 56 + 24$$

$$A = 122 \text{ ft}^2$$

(28) If a chunk of ice is in the shape of a rectangular prism with length 2 cm , height 1 cm , and width 3 cm , find its surface area.

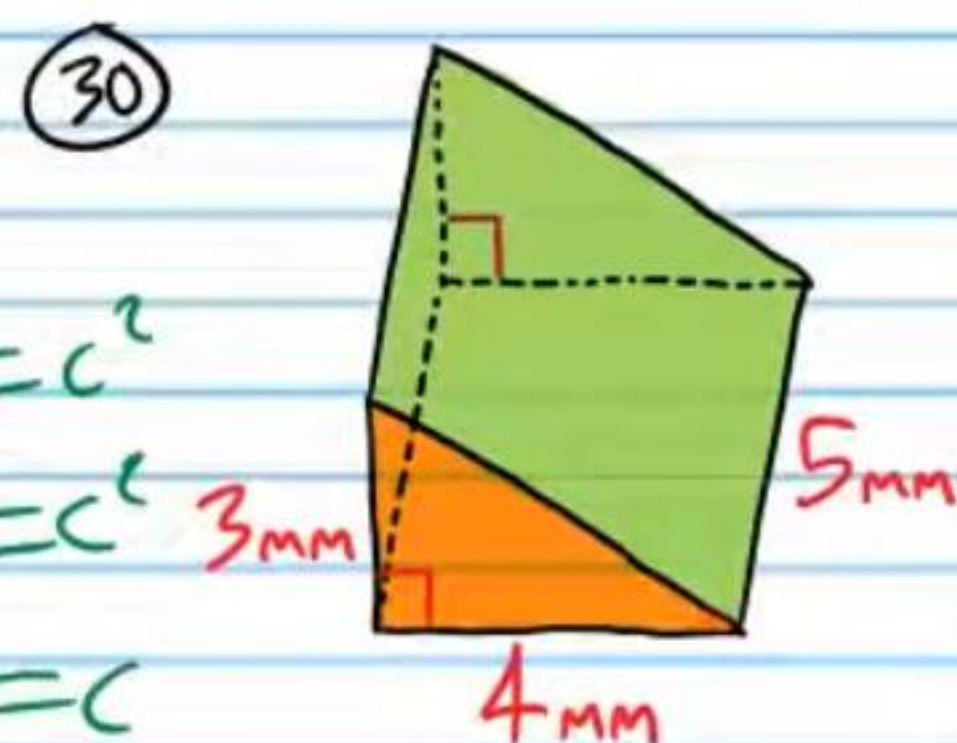
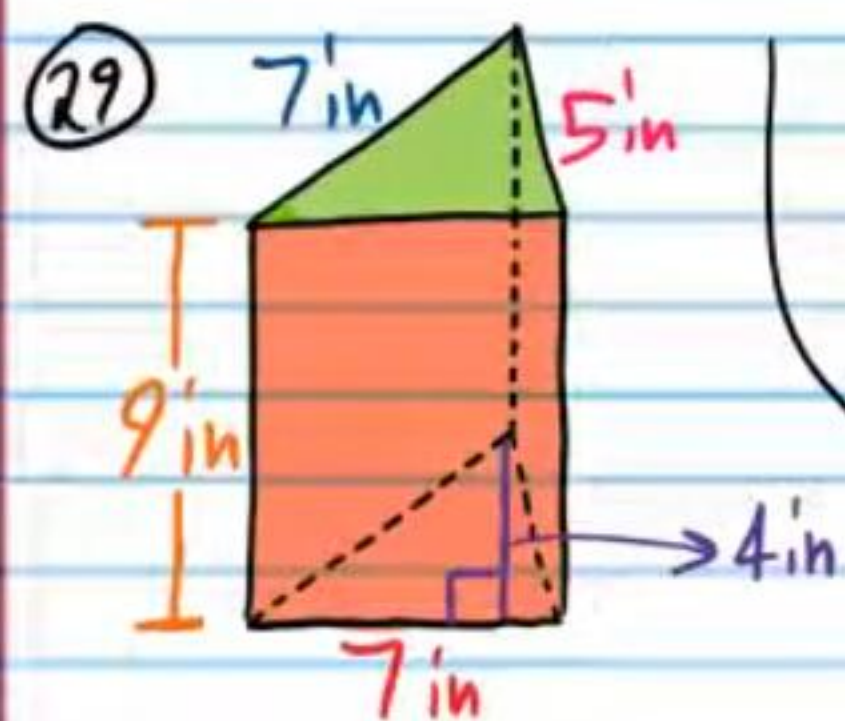


$$A = 1(2)(2) + 3(1)(2) + 3(2)(2)$$

$$A = 4 + 6 + 12$$

$$A = 22 \text{ cm}^2$$

Find the surface area of each prism below.



$$a^2 + b^2 = c^2$$

$$3^2 + 4^2 = c^2$$

$$5 = c$$

$$A_{\text{Tot}} = A_{\text{Tri}} + A_{\text{Rect}}$$

$$A_{\text{Tot}} = \frac{1}{2}(7)(4)(2) + 9(7) + 9(5) + 7(9)$$

$$A_{\text{Tot}} = 28 + 63 + 45 + 63$$

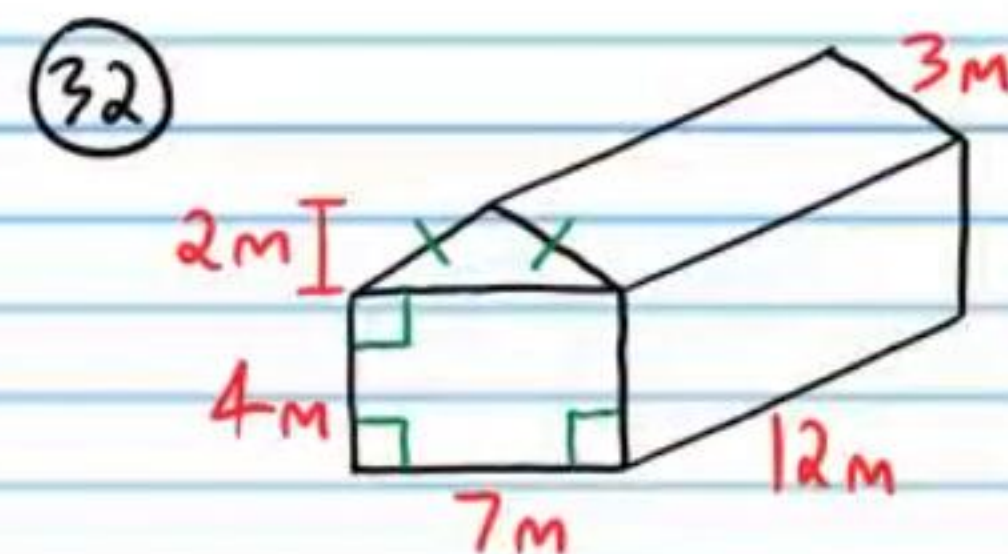
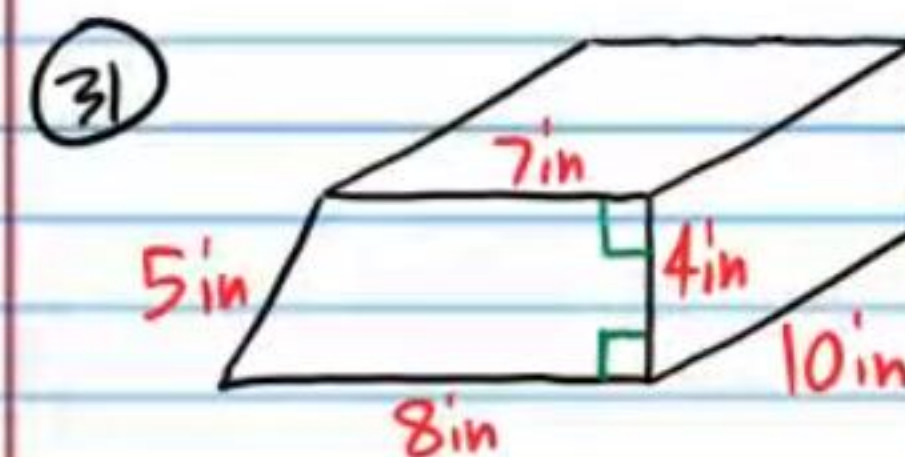
$$A_{\text{Tot}} = 199 \text{ in}^2$$

$$A_{\text{Tot}} = A_{\text{Tri}} + A_{\text{Rect}}$$

$$A_{\text{Tot}} = \frac{1}{2}(4)(3)(2) + 5(5) + 5(4) + 5(3)$$

$$A_{\text{Tot}} = 12 + 25 + 20 + 15$$

$$A_{\text{Tot}} = 72 \text{ mm}^2$$



$$A_{\text{Tot}} = A_{\text{Trap}} + A_{\text{Rect}}$$

$$A_{\text{Tot}} = \frac{4(7+8)}{2}(2) + 4(10) + 7(10) + 8(10) + 10(5)$$

$$A_{\text{Tot}} = 60 + 40 + 70 + 80 + 50$$

$$A_{\text{Tot}} = 300 \text{ in}^2$$

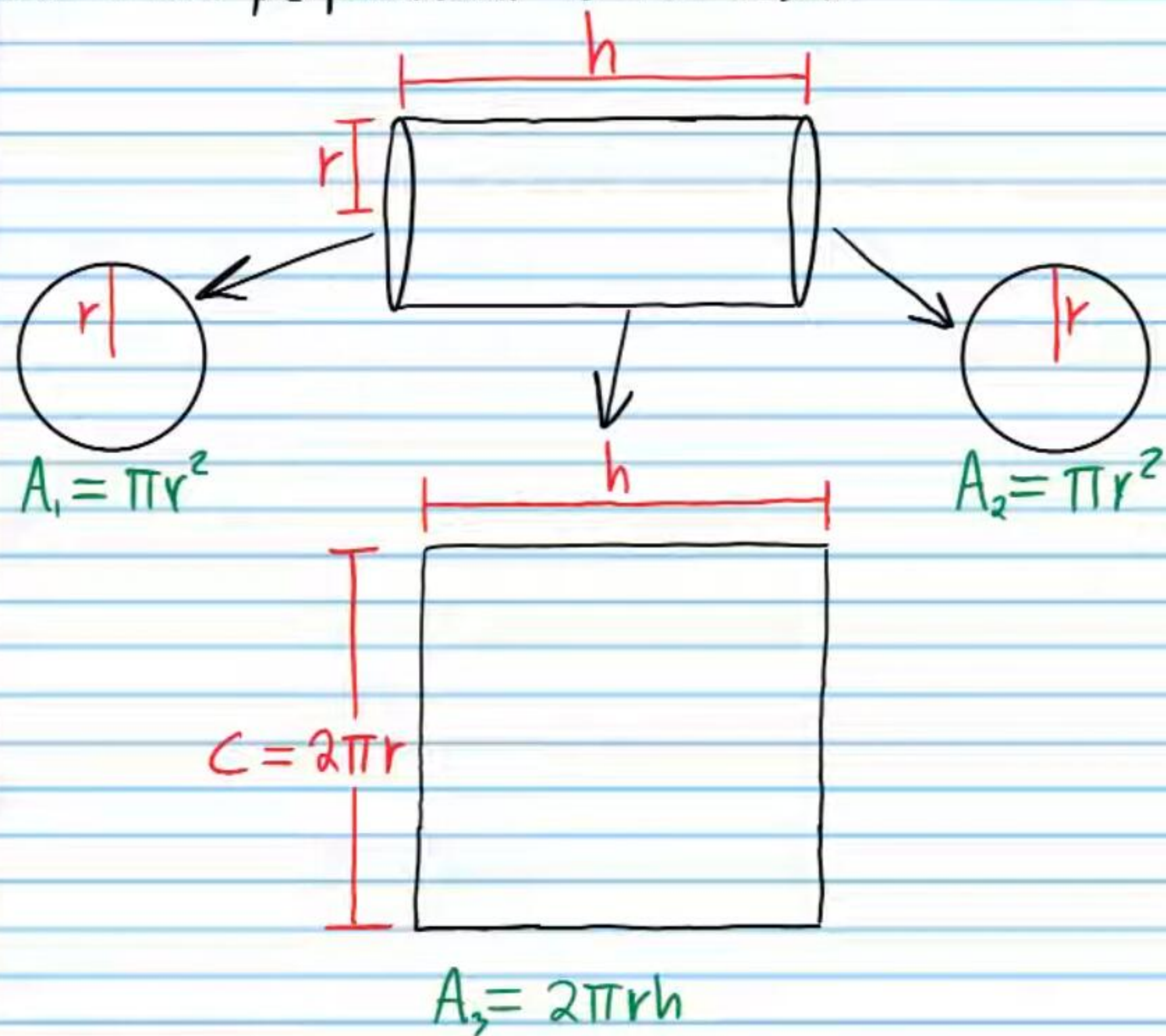
$$A_{\text{Tot}} = A_{\text{Tri}} + A_{\text{Rect}}$$

$$A_{\text{Tot}} = \frac{1}{2}(7)(2)(2) + (4)(7)(2) + 4(12)(2) + (7)(12) + 3(12)(2)$$

$$A_{\text{Tot}} = 14 + 56 + 96 + 84 + 72$$

$$A_{\text{Tot}} = 322 \text{ m}^2$$

Right Cylinder: A three dimensional shape with two congruent circular sides called **bases** which are parallel. The surface connecting the bases contains segments which are perpendicular to the bases.

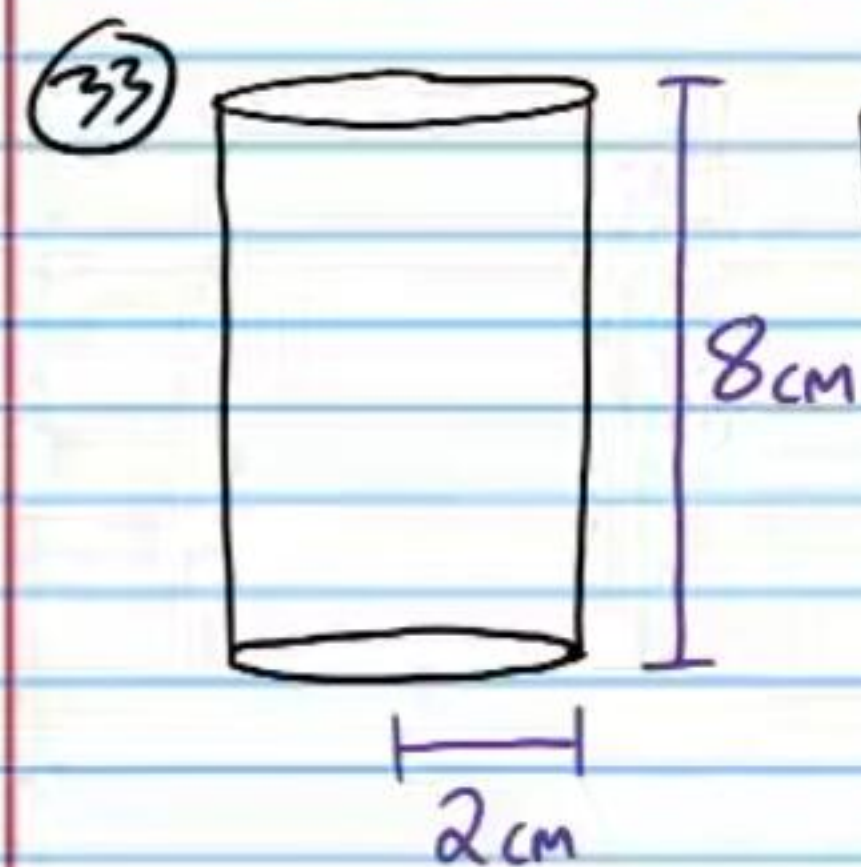


Surface Area of a Right Cylinder

$$A = 2\pi r^2 + 2\pi r h$$

$r = \text{radius} \ \& \ h = \text{height}$

Find the surface area of each right cylinder below.
Leave each answer as a multiple of π .

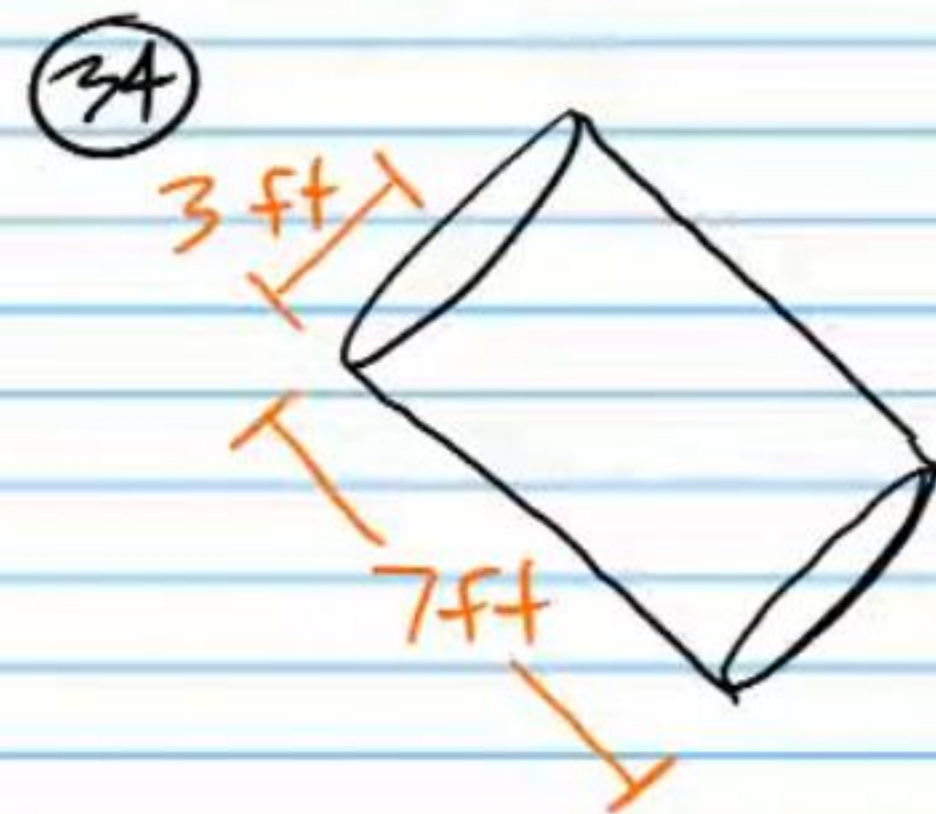


$$A_{\text{Tot}} = 2\pi(2)^2 + 2\pi(2)(8)$$

$$A_{\text{Tot}} = 2\pi(4) + 2\pi(16)$$

$$A_{\text{Tot}} = 8\pi + 32\pi$$

$$A_{\text{Tot}} = 40\pi \text{ cm}^2$$



$$A_{\text{Tot}} = 2\pi(3)^2 + 2\pi(3)(7)$$

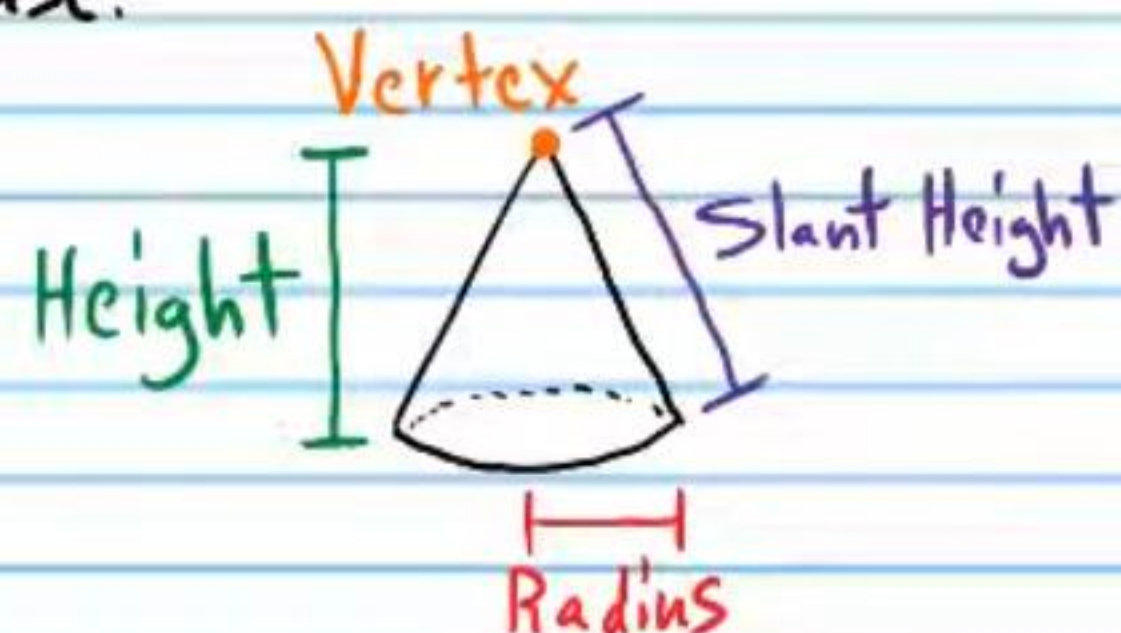
$$A_{\text{Tot}} = 2\pi(9) + 42\pi$$

$$A_{\text{Tot}} = 18\pi + 42\pi$$

$$A_{\text{Tot}} = 60\pi \text{ ft}^2$$

Right Cone: A three dimensional shape with a circular base and a curved surface converging at a point called the vertex (or apex). The segment joining the vertex and the center of the base is perpendicular to the radii of the base.

Example:

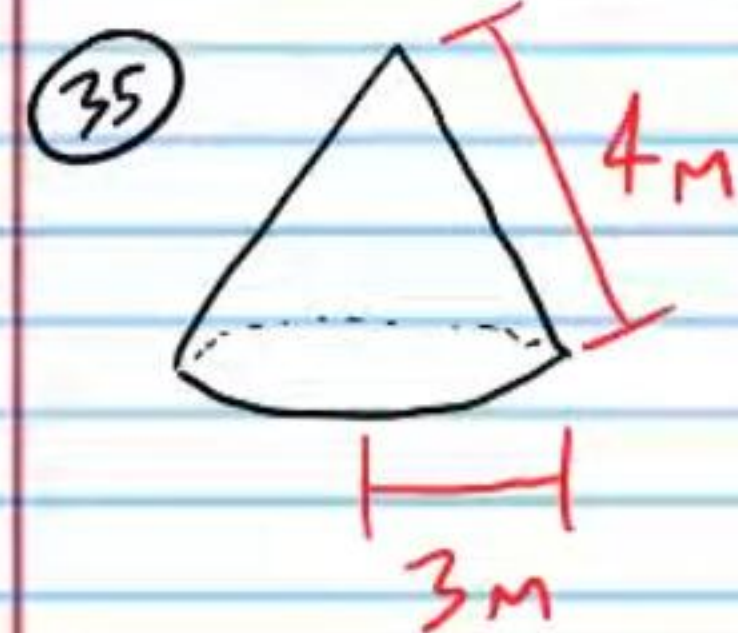


Surface Area of a Right Cone

$$A = \pi r^2 + \pi r L$$

r = radius of base & L = slant height

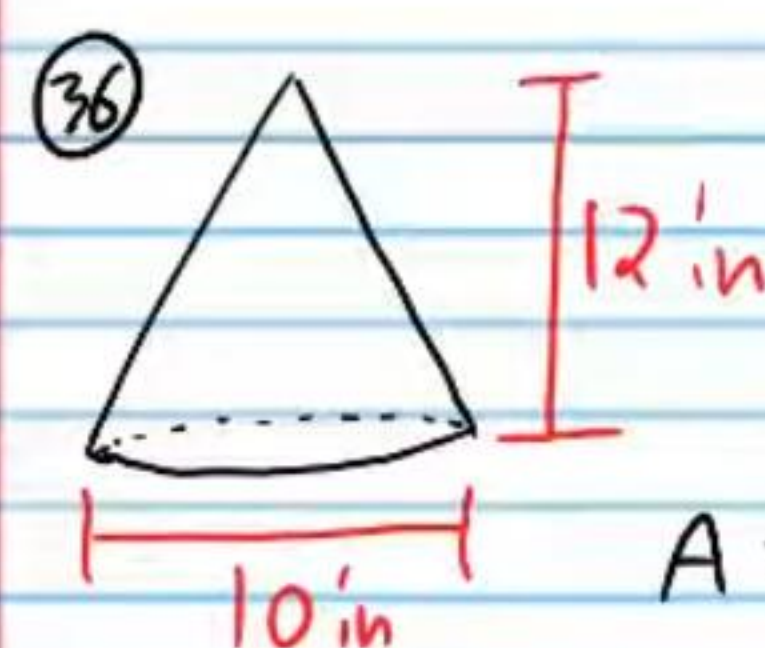
Find the surface area of each right cone below.
Leave each answer as a multiple of π .



$$A = \pi(3)^2 + \pi(3)(4)$$

$$A = 9\pi + 12\pi$$

$$A = 21\pi \text{ m}^2$$

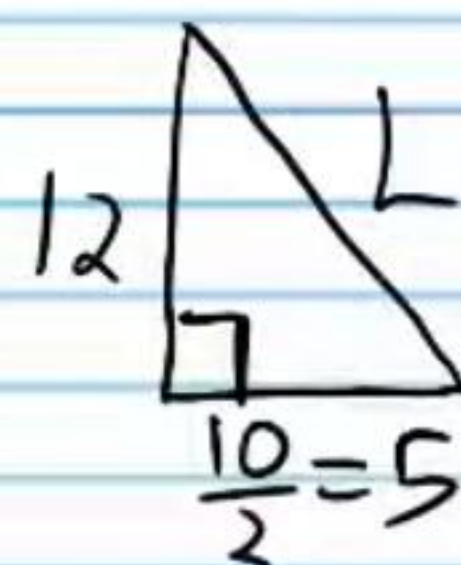


$$r = \frac{10}{2} = 5$$

$$A = \pi(5)^2 + \pi(5)(13)$$

$$A = 25\pi + 65\pi$$

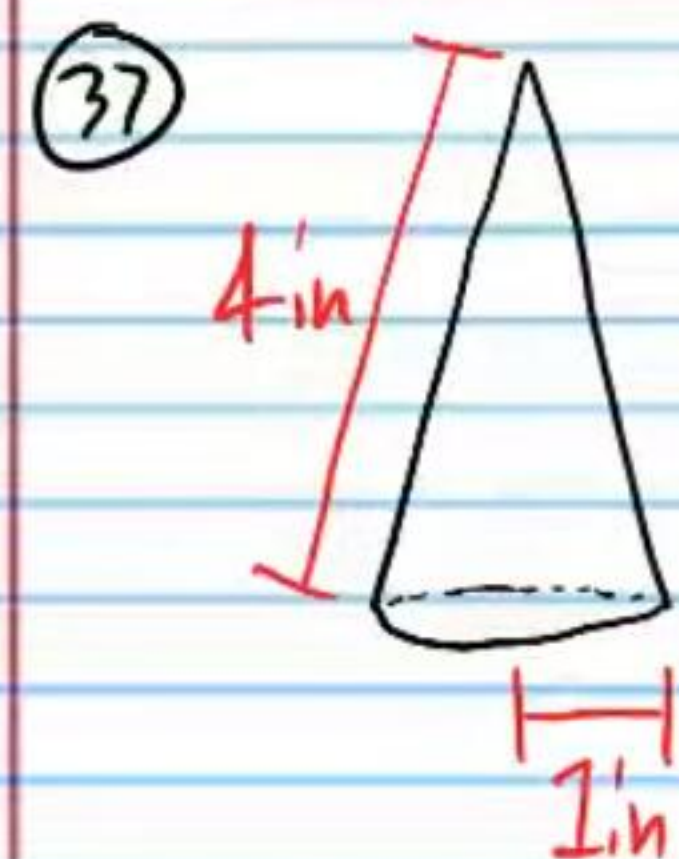
$$A = 90\pi \text{ in}^2$$



$$a^2 + b^2 = c^2$$

$$12^2 + 5^2 = L^2$$

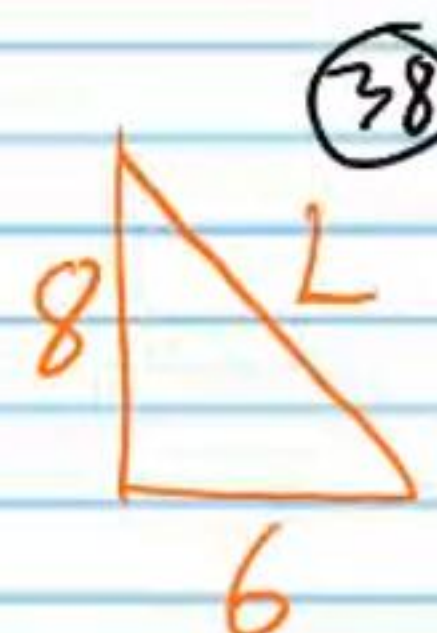
$$L = 13$$



$$A = \pi(1)^2 + \pi(1)(4)$$

$$A = \pi + 4\pi$$

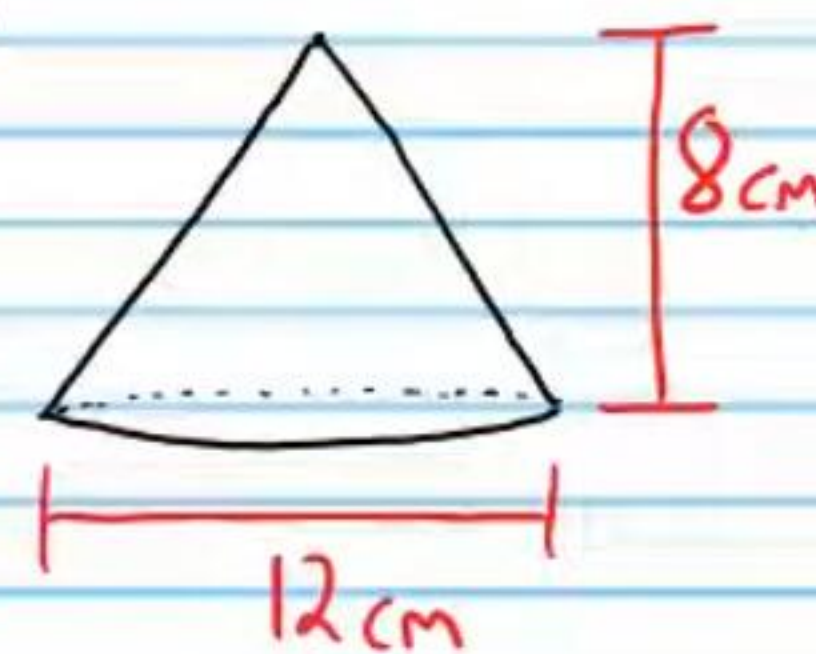
$$A = 5\pi \text{ in}^2$$



$$a^2 + b^2 = c^2$$

$$8^2 + 6^2 = L^2$$

$$L = 10$$



$$A = \pi\left(\frac{12}{2}\right)^2 + \pi\left(\frac{12}{2}\right)(10)$$

$$A = \pi(6)^2 + \pi(6)(10)$$

$$A = 36\pi + 60\pi$$

$$A = 96\pi \text{ cm}^2$$

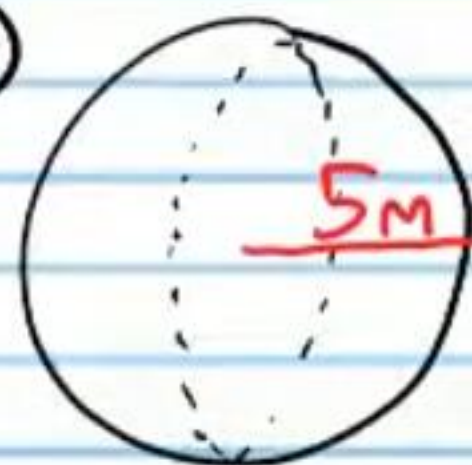
Surface Area of a Sphere

$$A = 4\pi r^2$$

$r = \text{radius}$

Find the surface area of each sphere below.
Leave each answer as a multiple of π .

39



$$A = 4\pi(5)^2$$

$$A = 4\pi(25)$$

$$A = 100\pi \text{ m}^2$$

40



$$A = 4\pi(3)^2$$

$$A = 4\pi(9)$$

$$A = 36\pi \text{ ft}^2$$